Grace Y. Mahoney

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TECHNICAL SKILLS

Programming Languages: C/C++, Python, MATLAB, HTML/CSS, ROS2, JavaScript, Typescript
Mechanical Design & Analysis: SolidWorks, Finite Element Analysis, stress analysis, torque calculations
Robotics & Controls: ROS2, Kinematics, PID Control, Motion Planning, Path Planning, Computer Vision
Hardware & Embedded Systems: Arduino, KiCad, sensor integration, embedded systems
Simulation & Development Tools: MuJoCo, Linux, Git

EDUCATION

Worcester Polytechnic Institute (WPI) - Worcester, MA | GPA: 3.8/4.0
Bachelor of Science in Robotics Engineering *Aug 2023 - May 2027*
Bachelor of Science in Computer Science *Aug 2023 - May 2027*
Dean's List Fall 2023, Spring 2024, Fall 2024, Fall 2025, Spring 2026
Relevant Course Work: Introduction to Robotics, Unified Robotics I: Mechanical Applications in Robotics, Unified Robotics II: Sensing and Perception in Robotics, Unified Robotics III: Manipulation, Control Engineering I, Embedded Systems for Robotics, Principles of Physics-Mechanics, Oscillations and Waves

PROJECTS

Symbiotic Multi-Agent Construction (SMAC) 7.0 | *Aug 2026 – May 2027*
- Developed a *MuJoCo-based simulation* of 5-DOF Inchworm robots and intelligent Smart Blocks with kinematic constraints, collision detection, and asynchronous communication.
- Developed *decentralized swarm algorithms* using stigmergic cell-weight gradients to guide navigation, material retrieval, collision avoidance, and autonomous construction from locally available information.

Autonomous TurtleBot Navigation with ROS2 | *Oct 2025 – Dec 2025*
- Implemented *A*-based path planning* with *LiDAR* obstacle detection and *AMCL* localization for autonomous navigation.
- Solved the "kidnapping problem," enabling the robot to re-localize and autonomously return to its home position using *ROS2 navigation* stack.

Romi 5-Bar Linkage with Sensor-Equipped End Effector | *Jan 2025 – Mar 2025*
- Designed and analyzed a *5-bar robotic linkage* and *custom gripper* using *MATLAB*, torque calculations, and *FEM stress analysis in SolidWorks* as a team of 4.
- Implemented *Arduino-based control* with *motors* and *sensors* to pick and place a box from 3 shelves, demonstrating *kinematic linkage control* and *sensor-driven activation*.

4-DOF Robotic Arm with Computer Vision for Color-Based Sorting | *Aug 2025 – Oct 2025*
- Explored *forward* and *inverse kinematics, trajectory planning,* and *Jacobian singularity* for precise *task-space* control for end-effector movement.
- Integrated *computer vision* for autonomous color-based ball sorting, demonstrating control, task planning, and kinematic linkage applications.

WORK EXPERIENCE

Dog Walker / Pet Sitter, Rover and Independent | *2016 - Present*
- Demonstrated reliability, time management, and accountability while managing recurring client schedules and responsibilities.

Suite Runner, Polar Park*, Worcester, MA* | *May 2026 – Present*
- Demonstrated organization, responsiveness, and teamwork while preparing event suites, coordinating catered food and supplies, supporting servers, and completing post-event cleanup.

ACTIVITIES

Treasurer, Autonomous Underwater Vehicle Team (AUV)
Public Relations Committee, Society of Asian Scientists and Engineers (SASE)